

A Cubic Separation Law for Wide Binaries: an Exponent With No Free Parameters

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Version 3 · 2026-07-30. The s^3 prediction, its five falsifiers and the frozen pre-registration are UNCHANGED. v3 (i) reduces the coefficient to a single rational, $a_0 = \kappa c \sqrt{G\rho_\Lambda}$ with $\kappa = 1/2$ (§1); (ii) tightens the ECCENTRIC-ORBIT error budget from a factor of several to $\lesssim 8\%$ using the action's own $\square_w$, and records two exact identities (§2.1); (iii) adds an independent CMB check of the a_0 footing, crediting Gnedin (2008) for posing and first deciding that fork (§8); and (iv) reports six closed classes in the search for what fixes κ — a negative result. Version 2 (2026-07-28) settled the branch ambiguity, generalised the exponent, and resolved feasibility negatively; Version 1 (2026-07-25) is superseded. Scripts listed in §8.

The claim, first

In wide binaries wider than about 50 kAU, the relative velocity should exceed the Newtonian value by an amount growing as the **cube of the separation**:

$$\gamma_v(s) - 1 \simeq \frac{1}{2} \left[\nu(y_{\text{ext}}) - 1 \right] \frac{\omega_c^2}{GM} s^3.$$

The **exponent is 3**. It is derived, not fitted, and it cannot be tuned. Every known contamination channel in wide-binary samples rises as $s^{1/2}$ instead. Exponent 3 against exponent 0.5 — that contrast is the measurement.

What would kill it. Any one of: no excess beyond ~50 kAU in a contamination-controlled sample; an excess whose log-slope is inconsistent with 3; a knee that does not move as $M^{1/3}$; an amplitude inconsistent with ω_c^2/GM ; or a boost $\gamma_v \approx 1.02$ found *inside* 30 kAU (**renumbered 2026-07-30 from 1.09 under the retired $\alpha = 1$ kernel; see the notice in §2**).

What it cannot do. It cannot be measured with Gaia DR3. We checked the catalogue ourselves; see §6.

1. Two scales, one of them free

The framework is *modified inertia* — the modification is in the matter sector, not in gravity. Its acceleration scale is tied to the cosmological constant:

$$a_0 = \frac{cH_\Lambda}{Z} = c^2 \sqrt{\frac{\Lambda}{32\pi}} = 9.36 \times 10^{-11} \text{ m s}^{-2}, \quad Z = \sqrt{32\pi/3} = 5.78881.$$

New in v3 — the coefficient is a single rational, and “ $32\pi/3$ ” is an artifact. Writing 32π as $8\pi/\kappa^2$ and substituting $\Lambda = 8\pi G\rho_\Lambda/c^2$, every π , the 32 and the 3 all cancel:

$$\boxed{a_0 = \kappa c \sqrt{G \rho_\Lambda}}, \quad \kappa = \frac{1}{2}.$$

So $32\pi/3$ is not a compound of three geometric factors — it is what one statement looks like after being routed through Λ and Einstein’s 8π . The framework’s entire dimensionless content is the single **rational** $\kappa = 1/2$, and the honest form of the open problem is *why* $\kappa = \frac{1}{2}$, not *why* $32\pi/3$. Two consequences worth recording. (i) The standing transcendence obstruction — “ Z carries a transcendental $\sqrt{\pi}$ while flavour data are algebraic” — is correct about Z , but that $\sqrt{\pi}$ belongs to **general relativity’s** 8π , not to this framework, whose own number is rational. (ii) κ remains **postulated**: an exhaustive sweep of the kernel’s own conditions (Herglotz sum rule, $K(0) = 0$, $\|K\| \leq 1$, retarded analyticity, branch point, thermal saturation against every forced kernel constant, and the location statistics of the spectral measure) found **none** that fixes it, because each is imposed at a scale-free point of the dimensionless axis $z = \square_u/a_0^2$ and is therefore invariant under $a_0 \rightarrow \lambda a_0$. Six classes closed; the value is not derived.

⚠️ KERNEL CHANGED 2026-07-30. This paper was written with $\nu(y) = \sqrt{1 + 1/y}$ ($\alpha = 1$). The framework has since adopted $\mu(x) = x/\sqrt{1 + x^2}$ ($\alpha = 2$, Milgrom 1983), because the $\alpha = 1$ tail implies a constant sunward anomaly $a_0/2$ that is **1279× the Earth 2σ ephemeris bound**. Two consequences for this paper, both stated rather than absorbed: **(i)** the identity $g_{\text{obs}}^2 - g_{\text{bar}}^2 = a_0 g_{\text{bar}}$ below is an $\alpha = 1$ identity and no longer holds — it is retained as attribution, not as a live relation; **(ii)** the ungated boost quoted throughout as $\gamma_v \approx 1.09$ becomes ≈ 1.02 (orientation-averaged 1.0246, range 1.0182–1.0350 over both a_0 footings and both frozen g_{ext} values), per Amendment 3 to PREREGISTRATION_DR4.md, filed in the open before DR4. **The paper’s central result is unaffected**: the cubic gate law, its $s^{1/2}$ -versus- s^3 contrast and the $M^{1/3}$ knee are properties of the gate, not of the tail exponent, and the gated prediction remains $\gamma_v \rightarrow 1.000$.

Credit, plainly stated. The interpolation $\nu(y) = \sqrt{1 + 1/y}$ and the identity $g_{\text{obs}}^2 - g_{\text{bar}}^2 = a_0 g_{\text{bar}}$ are Milgrom (1999, *Phys. Lett. A* **253**, 273), Eqs. (8)–(9), with coefficient $2cH_\Lambda$. Milgrom (1999, 2015) and Smolin (2017, *Phys. Rev. D* **96**, 083523) had already tied a_0 to Λ with a 2π coefficient. **No priority is claimed for the Λ tie.** What is ours: the coefficient cH_Λ/Z , the modified-inertia completion, and the prediction below.

The second scale is a response gate. The matter sector’s effective response is

$$K_{\text{eff}} = 1 - S \left(|a|/a_0 \right) G(\omega), \quad G(\omega) = \frac{1}{1 + i\omega/\omega_c}, \quad S \rightarrow \frac{a_0}{2g_N} \text{ (deep-Newtonian),}$$

with $\omega_c \in [1.782, 2.211] \times 10^{-14} \text{ s}^{-1}$. Every number below is carried on both cosmological footings ($a_0 = cH_\Lambda/Z$ and $a_0 = cH_0/Z$, differing by $\sim 21\%$).

Why a gate must exist. On the frequency axis $|K| = 1$ *exactly* for every $\omega > a_0/2c$: the kernel saturates and has no high-frequency roll-off. So K gives an inner-disk star and the Earth the **same** response. Since the solar system must be Newtonian and galaxies must not be, something frequency-dependent with a roll-off is required. The gate’s existence is forced, not assumed.

Why its scale is not. Stated against interest: ω_c is a **free parameter**. A dimensional census puts every intrinsic scale of the theory 3.2–5.6 decades away from it, and consistency alone (galaxies must survive, the solar-system monopole must not) brackets it only to ~ 3 orders of magnitude. So a_0 's value, Z , the sign $s = -1$, **and** ω_c are postulated, not derived. Version 1 did not say this clearly enough; it does now.

2. What frequency a binary presents to the gate

The kernel's argument is fixed by the action, not chosen. The committed matter term is

$$S_{\text{matter}} = -\frac{1}{2} \int d^4x \sqrt{-g} \rho_m [s u^\mu K(\square_u/a_0^2) u_\mu], \quad \square_u f = u^a \nabla_a (u^b \nabla_b f),$$

so $\square_u$ is the second derivative *along the worldline*. On a circular orbit u_μ is an eigenvector of it:

$$\square_u u_\mu = -\Omega^2 u_\mu \quad (\text{identically; verified symbolically}).$$

The argument is therefore $z = -(\Omega c/a_0)^2$ – the **orbital frequency**. The acceleration *magnitude* $|a| = \Omega^2 R$ genuinely is constant on a circular orbit, but the action never feeds $|a|$ to K .

This settles an ambiguity Version 1 was explicitly conditional on. Had the kernel responded to $|a|$, the gate would stay open and the prediction would be $\gamma_v \approx 1.02$ instead (1.09 under the retired $\alpha = 1$ kernel). It does not. The result below no longer rests on a choice.

2.1 New in v3 – eccentric orbits, and how sharp the prediction actually is

Real wide binaries are **not circular**, so §2's identity is the wrong tool for them on its own, and this is the honest soft spot of v1 and v2. Off circles the eigenvector relation fails and a *closure* must be chosen – a prescription for reducing $\square_u u_\mu$ to the scalar argument of K . The repo's prior verdict on that closure was **free, bounded**. Two things are new here.

(a) The freedom is much smaller than “free” suggests – by two orders of magnitude. The prior bracket was computed over time-weightings of $|a|^2$, which know nothing about $\square_u$. Using the action's actual $\square_u u$ instead collapses the spread. On radial infall onto $10^{12} M_\odot$ – the maximally non-circular case, i.e. a hard upper bound on eccentricity effects:

infall from	$ a ^2$ -weighting family	action $\square_u$ family	narrowing
300 kpc	$\nu = 1.170, 2.364, 7.837$ – spread 570%	$\nu = 1.0004, 1.0062, 1.0797$ – spread 7.9%	72×
100 kpc	spread 172%	spread 0.9%	189×

So the off-circular ambiguity in ν is $\lesssim 8\%$ in the worst case, not a factor of several. The s^3 prediction is correspondingly **sharp to that level for eccentric pairs**, which is what the Gaia DR4 shape test actually needs. This upgrades the prediction from quasi-linear to nearly sharp.

(b) Two exact identities, and an honest limit. Symbolically, at matched local acceleration a and radius r :

$$\text{circular: } \frac{\square_u u}{u} = -\Omega^2 = -\frac{a}{R}, \quad \text{radial infall: } \frac{\square_u u^r}{u^r} = \frac{2GM}{r^3} = +\frac{2a}{r}.$$

A factor 2 and an **opposite sign**. Stated against interest: the action does **not** eliminate the freedom, because $\square_u u$ is *not parallel* to u off circles — the spatial and time components differ by 8 to 5 orders along the infall — so a projection must still be chosen. The action narrows; it does not close.

3. The gate is shut in the window everyone measures

A pair of total mass M at separation s has $\Omega = (GM/s^3)^{1/2}$. For $M = 1.5 M_\odot$ at 10 kAU, $\Omega = 2.44 \times 10^{-13} \text{ s}^{-1}$ — **11× above the top of the whole ω_c window**. The gate is shut, $\text{Re } G = 0.005\text{--}0.008$, and $\gamma_v = 1.0004\text{--}1.0006$: within 0.04σ of Newton.

Two radii run in opposite directions with separation:

$$r_M = \sqrt{GM/a_0} \quad (\text{sub- } a_0 \text{ begins}), \quad r_{\text{gate}} = (GM/\omega_c^2)^{1/3} \quad (\text{gate opens}).$$

Since $\Omega \propto s^{-3/2}$ while $g \propto s^{-2}$, always $r_{\text{gate}} > r_M$ — the ratio is 4.54–7.76 across 0.5–3.0 $M_\odot$, both footings, both edges. **Between them a pair is already below a_0 but still gate-shut, so it must look Newtonian.** For 1.5 $M_\odot$ that dead zone runs ~10 to ~50–60 kAU.

Galaxies are untouched, which is what makes the gated reading admissible. Galactic orbital frequencies are $\Omega = v/r$, nine orders smaller. Across the Milky Way solar circle and outer disk, dwarfs, low-surface-brightness disks, massive spirals and the inner disk, the largest value found is $\Omega/\omega_c = 0.364$ ($\text{Re } G \geq 0.883$). Gate open in galaxies, shut in wide binaries: the radial acceleration relation and the baryonic Tully–Fisher relation are unaffected.

4. The cubic law — and why the exponent counts poles

Deep in the gate-shut regime $\Omega \gg \omega_c$, so $\text{Re } G \rightarrow (\omega_c/\Omega)^2$; substituting $\Omega = (GM/s^3)^{1/2}$ gives $\text{Re } G \rightarrow (\omega_c^2/GM) s^3$, hence the boxed law. The measured local slope $d \ln(\gamma_v - 1)/d \ln s$ is **3.00** at 5 kAU, falling through the knee to 0.06 once saturated.

An n -pole gate gives $|G_n|^2 \rightarrow (\omega_c/\Omega)^{2n}$, so

$$\gamma_v - 1 \propto s^{3n} \quad \Rightarrow \quad n = p/3.$$

Verified numerically at $n = 1, 2, 3$ (slopes 3.00, 6.00, 8.99). **The measured exponent returns the gate’s pole count.** The committed gate is one-pole and predicts exactly 3; 6 would mean two poles; anything not a multiple of 3 rules out the rational-pole form. (*The identity $|G|^2 = \text{Re } G$ used above holds only at $n = 1$; for $n > 1$ the law uses $|G_n|^2$.*)

The exponent is parameter-free even though the amplitude is not. ω_c is free (§1) and enters the amplitude — but it does **not** enter the exponent. That asymmetry is the whole reason this is worth measuring. Three mass powers follow from the same closed form and are recovered

numerically: $r_{\text{gate}} \propto M^{0.3333}$, $r_{\text{efe}} \propto M^{0.5000}$, fixed-separation excess $\propto M^{-0.9935}$ (asymptotic; knee-contaminated to $M^{-0.857}$ by 30 kAU, so fit the local slope rather than assuming a global $1/M$).

5. The discriminant

Every published contamination channel carries a *fixed velocity scale*, so its scaled relative velocity rises as $s^{1/2}$: chance alignment (El-Badry, Rix & Heintz 2021, via $R_{\text{chance align}}$), unresolved tertiaries (Peñarrubia 2021; Tyler, Green & Goodwin 2023), close-binary contamination. **Exponent 0.5 against 3.0.** Signal and background have different separation dependences, so a joint fit is well posed.

This matters because the *amplitude* axis is unusable: the two groups analysing the same public catalogue differ by $\Delta\gamma_v \approx 0.174$, about **2.1× the entire predicted boost**. No amplitude test survives that spread. A shape test does.

6. Why this waits for Gaia DR4 — now resolved, negatively

Version 1 said feasibility “cannot be settled without the catalogue in hand” and “may be fatal to a DR3 test.” We pulled the catalogue. It is fatal.

Because the excess grows as s^3 , nearly all the signal sits at large separation:

s [kAU]	γ_v	note
30	1.0125	the cut used by every published analysis
50	1.0392	—
100	1.0808	—
200	1.0932	catalogue’s reach

At the 30 kAU cut the predicted excess (1.2×10^{-2}) is an order of magnitude below the inter-group systematic — **the conventional window cannot test this**. And at our own recipe’s cuts ($d < 200$ pc, $R_{\text{chance align}} < 0.1$, RUWE < 1.4 both components, MS–MS), the El-Badry, Rix & Heintz (2021) Gaia eDR3 catalogue contains only

436/270/155/60/24/3 pairs across the 30 → 236 kAU bins,

i.e. **364 clean pairs beyond 50 kAU** and ~70 beyond 100 kAU. The shape test needs ~2000 per bin for 5σ separation of $p = 3$ from $p = 0.5$ (~400 gives only $\sim 2.3\sigma$). Relaxing the distance cut does not rescue it: proper-motion velocity error scales as (distance × pm error) while the sample requirement scales as σ^2 , so raw counts at $d < 1000$ pc are ~15× larger but the usable gain is far smaller — and even at face value only 2 of 6 bins clear 2000, while a log-slope fit needs most bins populated.

Ground-based astrometry cannot substitute. The test needs $\sigma_{\text{pm}} \lesssim 0.02$ mas yr^{−1} at 200 pc; Rubin/LSST delivers ~0.1–1.0 over ten years. **Gaia DR4 is the only route.**

7. Pre-registration

Since the test cannot run until DR4, the prediction is frozen in the open beforehand. The pre-registration (prep_2026/gaia_dr4_prep/PREREGISTRATION_DR4.md, Amendment 1, 2026-07-27, SHA-256 a309a502332b84ad521385b1c2031652849c9ce358c396f680f2308dd5ba1c13) records:

- the frozen prediction $p = 3$ for the log-slope beyond r_{gate} ;
- $\gamma_v = 1.0004\text{--}1.0006$ inside 2–30 kAU, i.e. **a Newtonian DR4 result in that window confirms this branch rather than refuting it** — which the unamended pre-registration would have scored as a kill;
- the five falsifiers listed at the top of this note;
- and that DR4 γ_v constrains the ν +EFE+gate prescription, **not** the value of a_0 or Z . No outcome may be reported as measuring either.

8. What is not claimed

- **Not a test of a_0 or Z .** A confirmed s^3 rise establishes a frequency scale ω_c and a one-pole response. $Z = 5.7888$ and the conventional $2\pi = 6.2832$ differ by 7.87%, both lie inside the $\pm 16\%$ empirical a_0 box, and no arena examined here separates them. Against interest: matching Chae (2024b)’s central γ on this framework’s own kernel needs $a_0 \approx 1.9 \times \text{canonical}$, which at face value disfavors the Λ -tied coefficient specifically.
- **Nothing about whether dark matter exists.** A wide-binary result constrains the weak-field *force law*, not the matter content. $\gamma_v = 1$ is Newton’s prediction — shared by Λ CDM, by this framework’s own gated branch, and by modified-force-plus-particle hybrids.
- **a_0 ’s value, κ (equivalently Z), the sign $s = -1$ and ω_c are postulated, not derived.** New in v3: six independent classes of condition on the kernel’s own spectral structure were swept for one that would fix κ , and all six are closed — every committed condition sits at a scale-free point of $z = \square_u/a_0^2$ and is invariant under $a_0 \rightarrow \lambda a_0$.

New in v3 — the footing this paper carries is the one independently favoured, and credit for that question is not ours. The paper runs both a_0 footings throughout. The fork itself — whether a_0 couples to $cH(z)$ or to the dark-energy density $\sqrt{8\pi G\rho_\Lambda/3}$ — was posed and tested long before this work: **Gnedin (2008)**, *The Redshift Evolution of the Tully–Fisher Relation as a Test of Modified Gravity* (arXiv:0809.2790), states both options explicitly, tests them against high- z Tully–Fisher data to $z = 1.2$, and concludes that dark-energy coupling can be consistent while Hubble coupling requires unreasonable w — “*this would appear to favor coupling to the dark energy density.*” See also arXiv:0802.1526 on whether G and a_0 vary with epoch.

We add only an **independent check from a different epoch and different systematics**, which is worth recording because Gnedin explicitly flagged possible unaccounted systematics. Applying the two footings to the recombination plasma: since the acoustic acceleration is $a_{\text{ac}} \sim c_s H$, a rising $a_0 = cH(z)/Z$ makes H **cancel**, leaving $y = Z/\sqrt{3(1+R)} \approx 2.6\text{--}3.3$ at every epoch — so $\nu - 1 \approx 15\%$ throughout the entire acoustic era, not as a late correction. Propagated through $c_s^2 = c^2/[3(1+R)]$ over CAMB’s exact background (our sound-horizon integral reproduces CAMB’s own $r_\star = 144.44$ Mpc to 0.00%), this shifts $\theta_\star$ by $\approx 1.5\%$ against Planck’s 0.03% measurement — excluded. The constant, ρ_Λ -tied footing shifts $\theta_\star$ by 3×10^{-7} and is consistent. **Same verdict**

as Gnedin (2008), from the CMB rather than from Tully–Fisher. Stated as a limitation: this is a background-plus-recombination treatment, not a modification of the Boltzmann perturbation hierarchy, and a fully relativistic MOND theory (AeST; Skordis & Złośnik) already reproduces the CMB — so the rigorous form of this question belongs to that theory, not to an $a_0(z)$ scaling argument.

Reproducibility. Every number is printed by committed scripts, all exit 0, no hard-coded verdicts, both footings and both ω_c edges throughout: `mi_dcac_branch_settled_2026.py` (5/5, §2), `mi_wb_gate_fork_2026.py` (11/11, §3), `mi_wb_cubic_rise_2026.py` (10/10, §4), `mi_wb_exponent_pipeline_2026.py` (6/6, §4–5), `mi_wb_dr3_feasibility_2026.py` (5/5, §6), `mi_omegac_anchor_2026.py` (8/8, §1), `count_wb_elbadry2021.py` (catalogue counts). New in v3: `mi_kappa_spectral_reduction_2026.py` (§1, the kappa reduction), `mi_offcircular_closure_collapse_2026.py` (§2.1, the closure narrowing and the two identities), `mi_three_classes_2026.py` + `mi_thermal_class_nogo_2026.py` + `mi_bootstrap_circularity_2026.py` (§8, the six closed classes), `mi_cmb_camb_run_2026.py` (§8, the CAMB footing check). Repository: <https://github.com/carlzimmerman/zimmerman-formula>

Corrections carried from Version 1, recorded rather than quietly fixed: the 200-vs-30 kAU excess ratio is $7\times$, not the $>10\times$ first stated; the fixed-separation mass scaling is asymptotic ($M^{-0.9935}$ at 10 kAU) and knee-contaminated to $M^{-0.857}$ by 30 kAU; the operative sample requirement is the shape test’s ~ 2000 pairs/bin, not the ~ 265 implied by excess-detection alone; and ω_c is now declared a free parameter rather than described as a constrained window.

New in v3, and stated plainly: nothing in this version changes the s^3 prediction, its five falsifiers, or the frozen pre-registration. v3 sharpens the *error budget* (§2.1: eccentric-orbit closure now $\lesssim 8\%$ instead of a factor of several), simplifies the *coefficient* (§1: one rational $\kappa = 1/2$), and adds an *independently-credited* footing check (§8). No new observable, no new claim, and one additional negative result — six closed classes in the search for what fixes κ .

References

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